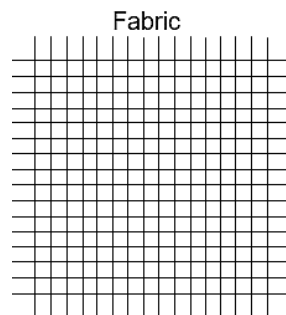


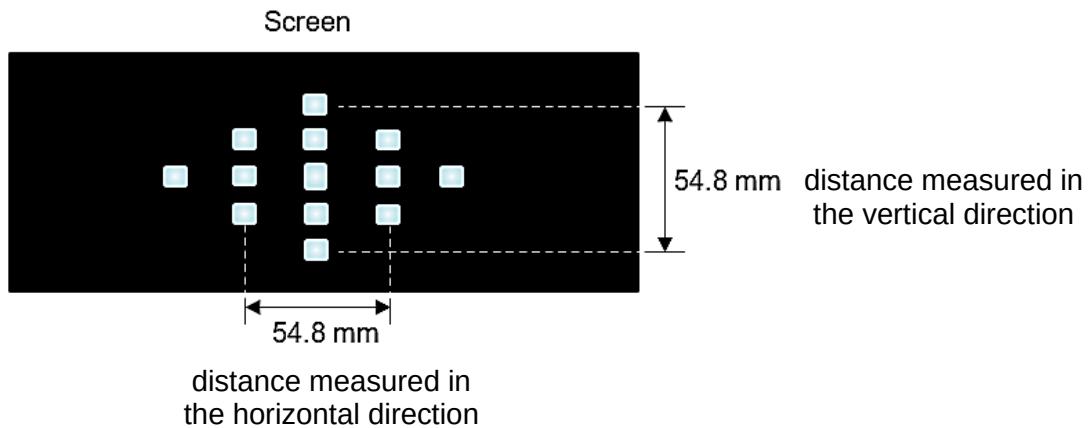
18

A fabric consists of closely-spaced horizontal and vertical threads as shown.



When a monochromatic light of wavelength 685 nm is incident normally on the fabric, a diffraction pattern is observed on a screen placed at a distance of 2.00 m away, as shown below.

[Turn over



The separation between the horizontal threads is determined by the vertical fringe separation.

What is the separation between the horizontal threads of the fabric?

- | | | | | | | | |
|----------|---------------------------------|----------|---------------------------------|----------|---------------------------------|----------|---------------------------------|
| A | $2.50 \times 10^{-5} \text{ m}$ | B | $5.00 \times 10^{-5} \text{ m}$ | C | $1.00 \times 10^{-4} \text{ m}$ | D | $9.48 \times 10^{-4} \text{ m}$ |
|----------|---------------------------------|----------|---------------------------------|----------|---------------------------------|----------|---------------------------------|

L3 Answer: C

Separation between the horizontal threads is determined by the vertical fringe separation.

Diffraction grating equation

$$d \sin \theta = n\lambda$$

Where

$$\theta = \tan^{-1} [(0.0548/2) / 2.00] = 0.7849^\circ$$

$$n = 2$$

$$\lambda = 685 \times 10^{-9}$$

Thus,

$$d = \underline{1.00 \times 10^{-4} \text{ m}}$$

Extension Question: For comparison, the spacing of the vertical threads is determined by the horizontal fringe separation:

$$d \sin \theta = n\lambda$$

Where

$$\theta = \tan^{-1} [(0.0548/2) / 2.00] = 0.7849^\circ$$

$$n = 1$$

$$\lambda = 685 \times 10^{-9}$$

Thus,

$$d = 5.00 \times 10^{-5} \text{ m}$$